Class 18: Fermi-Dirac Statistics: Part 2

In the previous class we looked at the process by which we expect to find the probability of occupancy of state at energy level ε_i, for a collection of fermions. Stated in words, the general equation we arrived at is indicated in Figure 18.1 below.

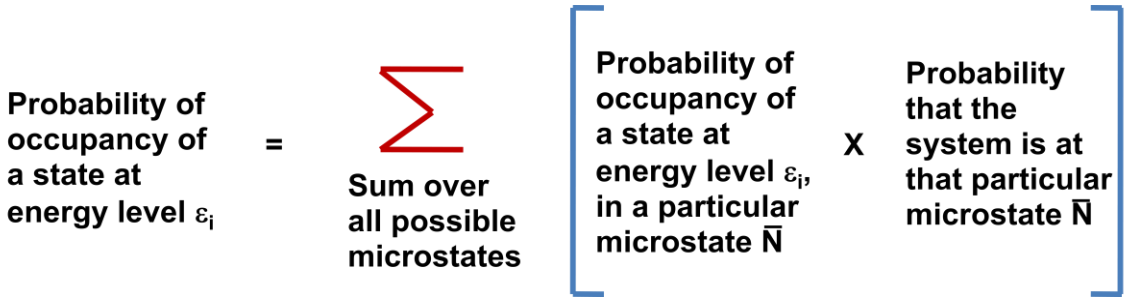


Figure 18.1: The general equation that enables us to determine the probability of occupancy of state at energy level ε_i, for a collection of fermions

The equation corresponding to Figure 18.1 above is:

$$f(\varepsilon) = \sum_{\bar{N}} f(\varepsilon|\bar{N}) \times p(\bar{N})$$

Written more elaborately, we noted that

$$f(\varepsilon) = \frac{1}{\omega_{total}} \sum_{\bar{N}} f(\varepsilon|\bar{N}) \times \omega(\bar{N})$$

Based on the approximation that statistical mechanics operates on, which is that in systems such as these, there exists a most probable microstate, $\bar{N}_0$, which is more probable than all of the other microstates combined, we have:

$$\omega_{total} = \omega(\bar{N}_1) + \omega(\bar{N}_2) + \omega(\bar{N}_3) + \cdots \approx \omega(\bar{N}_0)$$

Therefore

$$f(\varepsilon) \approx f(\varepsilon|\bar{N}_0)$$

In other words, our problem reduces to determining $\frac{n_i}{s_i}$, when $\bar{N} = \bar{N}_0$. We determine this $\frac{n_i}{s_i}$, by *partially* going over the math to determine $\omega(\bar{N}_0)$, during which we obtain an expression for $\frac{n_i}{s_i}$, when $\bar{N} = \bar{N}_0$. This is the direction we will proceed in, in the calculations below.

$$\omega(\overline{N}) = \frac{s_1!}{n_1!\,(s_1 - n_1)!} \times \frac{s_2!}{n_2!\,(s_2 - n_2)!} \times \ldots = \prod_i \frac{s_i!}{n_i!\,(s_i - n_i)!}$$

Since the function $\omega(\overline{N})$ and the function $\ln \omega(\overline{N})$, either both increase or both decrease, their maxima occur at the same point $\overline{N}_o$. Therefore instead of maximizing $\omega(\overline{N})$, we can chose to maximize $\ln \omega(\overline{N})$, which is mathematically easier to handle, and arrive at the same desired result, $\overline{N}_o$.

$$\ln \omega(\overline{N}) = \sum_i \ln s_i! - \sum_i \ln n_i! - \sum_i \ln(s_i - n_i)!$$

We can now make use of Stirling's approximation that for large X

$$\ln X! = X \ln X - X$$

Therefore,

$$\ln \omega(\overline{N}) =$$
$$\sum_i \{s_i \ln s_i - s_i - (n_i \ln n_i - n_i) - [(s_i - n_i)\ln(s_i - n_i) - (s_i - n_i)]\}$$

Simplifying

$$\ln \omega(\overline{N}) = \sum_i s_i \ln s_i - \sum_i n_i \ln n_i - \sum_i (s_i - n_i)\ln(s_i - n_i)$$

When $\ln \omega(\overline{N})$ reaches a maximum, $\partial \ln \omega(\overline{N}) = 0$

Since s_i are fixed, $\partial \ln \omega(\overline{N}) = 0$ implies

$$-\sum_i [\ln n_i - \ln(s_i - n_i)]\partial n_i = 0$$

Or $$-\sum_i \ln\left(\frac{n_i}{s_i - n_i}\right)\partial n_i = 0$$

In addition, the total number of particles present in the system is fixed, therefore, even if we move a few particles from one energy level to another, the total sum of those exchanges must be zero.

Hence:

$$\sum_i \partial n_i = 0$$

Similarly the total energy of the system is fixed, therefore even if we move a few particles from one energy level to another the total energy change corresponding to those exchanges, must be zero.

$$\sum_i \varepsilon_i \partial n_i = 0$$

Using the Lagrange method of undetermined multipliers, discussed in greater detail in Class 12, we obtain

$$-\ln\left(\frac{s_i - n_i}{n_i}\right) + \alpha + \beta\varepsilon_i = 0$$

Note: The negative sign in the first term in the equation above, is inserted by convention, to make the final results easier to interpret. Since the equations leading up to the above are all equal to zero, introduction of a negative sign before summing up is acceptable.

Simplifying further,

$$\frac{s_i}{n_i} - 1 = e^{\alpha + \beta\varepsilon_i}$$

Therefore:

$$\frac{n_i}{s_i} = \frac{1}{1 + e^{\alpha + \beta\varepsilon_i}} = f(\varepsilon_i)$$

It is relevant to restate here that the above expression for $f(\varepsilon_i)$, or $f(\varepsilon)$ in general, has appeared during our efforts to maximize $\omega(\overline{N})$. However, we take advantage of the underlying concept of statistical mechanics, that this expression for $f(\varepsilon)$ will be significantly greater in magnitude than all other possible contributions to $\frac{n_i}{s_i}$. Therefore we consider it justifiable, and as a very reasonable approximation, to treat this expression for $f(\varepsilon)$, as the overall probability of occupancy of a state at energy level ε, for a system subject to the constraints we have imposed. This expression for $f(\varepsilon)$ is then referred to as the Fermi-Dirac distribution. Only the values of the constants α and β remain to be specified. For the systems of interest to us, the constants α and β work out such that:

$$f(\varepsilon) = \frac{1}{1 + e^{\left[\frac{(\varepsilon - E_f)}{k_b T}\right]}}$$

where E_f is referred to as the Fermi energy, and will be elaborated on later. T is the absolute temperature, and k_b is the Boltzmann constant.

In summary, electrons qualify as Fermions, and we have derived the Fermi-Dirac distribution, which applies to a collection of fermions in thermal equilibrium. In the next class we will examine the features of the Fermi-Dirac distribution and see what these features imply.